Phase 3 — Semantic Deduplication (Cross-Source Same-Story Detection)

Thesis priority: #2 (cross-source duplicate detection) from §5.5 Future Works. **Date:** 2026-10-06 (FYP 2, Week 1)
Status: implemented and evaluated against the production corpus; default threshold 0.85; human-labeled validation added 2026-10-12 (§3.3).

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1. Problem statement (from FYP 1 evaluation, thesis §5.4)

Token/Jaccard similarity can fail when different publishers describe the same incident with different wording, causing repeat notifications.

Concrete FYP 1 weakness (`src/core/services.py`): the cross-source story matcher compares token sets (Jaccard) over title / body / AI summary. When two publishers use different words for the same incident — synonyms, different sentence structures, or a different language (English vs Bahasa Malaysia, extremely common in Sarawak sources) — token overlap collapses and both versions are delivered as "new" stories.

2. Changes made

2.1 Design

The FYP 2 vector table (`article_embeddings`, Phase 2) already stores one embedding per article over `title + ai_summary + category + location`. Same-story articles embed close together even when their wording differs, so the semantic check needs **no new infrastructure and no LLM calls** — it is pure SQL + in-memory cosine between already -stored vectors.

The semantic pass is an **addition to dedup layer 2**, not a replacement (AGENTS.md preserves three dedup layers):

1. unique-link check (unchanged),
2. cross-source story matching: Jaccard pass first (cheap, exact-wording duplicates), then the new cosine pass on pairs Jaccard cleared,
3. per-user `UserArticleDelivery` (unchanged).

Scope decision: the cosine pass runs at **delivery-time clustering** (`_cluster_ranked_articles_cross_source`), where articles have DB ids and (usually) vectors. Ingestion-time RSS-item dedup (`_deduplicate_items`) is left Jaccard-only: items are not yet in the DB, and embedding raw items would cost Ollama calls per item. Duplicate DB rows still get *clustered* before any user sees them, which is what prevents repeat notifications.

2.2 Config (`src/core/config.py`)

Variable	Default	Purpose
`CROSS_SOURCE_DEDUP_SEMANTIC_ENABLED`	`true`	Master switch; `false` = Jaccard-only (exact FYP 1 behavior).
`CROSS_SOURCE_DEDUP_SEMANTIC_COSINE_THRESHOLD`	`0.85`	Same-story floor, set from the labeled-pair eval below.

2.3 Code

- `src/ai/retriever.py`:
- `fetch_stored_vectors(article_ids)` — one batched SQL fetch of embedding rows; returns `None` on any failure so callers skip the semantic pass (graceful degradation, rule #8). pgvector text literals are coerced defensively (`_coerce_vector`) — the pooler returns vectors as `'[f,f,…]'` text without a type codec.
- `cosine_between(a, b)` — pure-python cosine.

`src/core/services.py` — `_cluster_ranked_articles_cross_source`: vectors for the batch are fetched once up front; after the Jaccard matcher clears a pair, cosine against each existing cluster representative ≥ threshold groups the article with `match_by="semantic"` (visible in `CROSS_SOURCE_DEDUP_DEBUG` logs). Articles without embedding rows are still clustered by Jaccard alone.

No schema change → no migration (change rule #2 satisfied trivially).

3. Evaluation (2026-10-06, production Supabase corpus)

`scripts/eval_semantic_dedup.py`; full output `benchmark/dedup_semantic_20261006.txt`, pair data `benchmark/dedup_pair_cosines.csv`.

3.1 Threshold evidence (249 labeled pairs, 45-day corpus, 1,739 embedded articles)

Pair set	n	min	p25	median	p75	max
`jaccard_caught` (Jaccard would flag)	40	0.755	0.870	0.927	0.965	1.000
`hard_positive` (same location, ≤48h apart, Jaccard clears)	149	0.570	0.675	0.707	0.774	0.957
`negative` (different location or >7d apart)	60	0.492	0.625	0.667	0.688	0.765

Reading: true same-story pairs form a high-cosine band (the `jaccard_caught` row and the top of the `hard_positive` range); negatives stay ≤ 0.77. **0.85 sits in the gap** — above every sampled negative and below the clear same-story cluster. It deliberately keeps the gray band (0.77–0.85) unmerged: those pairs are ambiguous, and an unmerged possible duplicate is a far cheaper error than a false merge that hides a distinct story.

3.2 Before/after simulation (articles from the last 14 days, production clustering code)

	Clusters
Jaccard only (FYP 1)	312
Jaccard + semantic	249
Merges made **only** by the semantic pass	71

Manual review of the 30 sampled semantic-only merges (~23% estimated false-merge rate): the true merges include **cross-lingual pairs Jaccard can never catch** — e.g. "Underground gas leak prompts shutdown at Kuala Baram oil palm site" ↔ "Gas bawah tanah bocor, ladang sawit ditutup sementara" (0.943), the Long Lellang helicopter-crash widow follow-ups (0.869), IPU/haze Malay↔English updates (0.856–0.864), utility repair updates (0.948–0.962), and the Datuk Song Swee Guan obituary cluster (0.953–0.977). Typical false merges are same-genre event roundups ("Sarawak Borneo Craft Festival targets RM3 mln in sales" ↔ "BCG 2.0 targets 30,000 visitors", 0.885).

Cost of a false merge is **soft**: clustering controls what a digest displays; merged members still contribute their source links to the cluster's "Sources:" line, so the reader can reach the suppressed article, but its title/summary are not shown.

3.3 Human-labeled validation (2026-10-12, 40 pairs, owner-labeled)

`benchmark/dedup_tuning_set_20261012.csv` (via `scripts/export_dedup_tuning_set.py` + quiz UI `scripts/label_dedup_quiz.py`): 30 pairs from the hard stratum (same location, ≤48h, cross-source) closest to the 0.85 boundary + 5 positive / 5 negative anchors.

Stratum	n	same_story	different_story	unsure
band (0.65–0.97, closest-to-0.85 first)	30	9	19	2
— of which cosine ≥ 0.85 (would merge)	21	6	13	2
anchor_positive (≥ 0.95)	5	4	1	0
anchor_negative (≤ 0.60)	5	0	5	0

Findings:

1. **The decision boundary is genuinely hard.** On this deliberately-hardest stratum, pairs at/above the 0.85 threshold are mostly *different* stories (13/19 excluding unsure → ~32% merge precision here). True same-story pairs sit interleaved at 0.849–0.851 (e.g. P03, the Jalan Dudong crash, cos 0.8499). No single threshold separates the two groups — raising it would break these true merges *and* the cross-lingual true merges from §3.1 (0.856–0.869).
2. **The dominant failure mode is exactly "same place, different incident."** The labeled different-story pairs are overwhelmingly same-location recurring coverage: day-after-day haze API roundups, same-road crashes on different days. Recurring events at one location embed close together (the embedding input includes the location text plus same genre/entities), so cosine alone cannot tell "same incident re-reported" from "new incident, same place." A second gray-zone signal (shared distinctive token / specific numeric value, per the mitigation in §5) is now the recommended next step rather than a threshold move.
3. **A labeling-policy subtlety surfaced:** several pairs are the *same underlying event* but one article frames it locally and the other nationally ("A focus in Sarawak only, B nationwide"). The owner labeled these different_story, i.e. the product policy treats geographic scope as decisive, not just the underlying event. This policy should be written into the report's dedup definition.
4. One anchor exception: P32 (cos 0.9561, day-after-day Kanowit API updates) was labeled different_story — recurring-event roundups can false-merge even well above the threshold.
5. **Caveat:** the band stratum is enriched for hard cases and is *not* representative of all pairs the system merges; the §3.2 simulation merge precision (77% true) and this stratum's 32% are measuring different populations. The honest report claim is: merges are reliable for one-off incidents, unreliable for recurring-location events.

3.4 Second-signal fix (2026-10-12)

Implemented in `services.py` (`_evidence_tokens_for_story_dedup`, `_shared_distinctive_evidence`) with config in `src/core/config.py`:

Variable	Default	Purpose
`CROSS_SOURCE_DEDUP_SEMANTIC_GRAY_ZONE_CEILING`	0.95	≥ ceiling: cosine merges directly (clear same-story). Below: second signal required.
`CROSS_SOURCE_DEDUP_SEMANTIC_SECOND_SIGNAL_ENABLED`	true	Master switch for the gray-zone gate.
`CROSS_SOURCE_DEDUP_SEMANTIC_RARE_TOKEN_MAX_DF`	3	A content token is "distinctive" if it appears in ≤ this many batch articles.

Rule: merge iff cosine ≥ 0.85 **and** (cosine ≥ 0.95 **or** the pair shares distinctive evidence — an incident-grade numeric token, or a content token with batch df ≤ 3). Numeric hygiene excludes times ("8am"), day ordinals ("22nd") and bare years: the labeling showed those fire on any same-day pair regardless of story.

Replayed against the 40 human labels (`scripts/eval_second_signal.py`):

Rule	Precision	Recall	F1
cosine ≥ 0.85 (before)	0.417	0.769	0.541
+ second signal, rare cap 3	**0.692**	0.692	**0.692**
+ second signal, rare cap 4	0.588	0.769	0.667
+ second signal, rare cap 5	0.476	0.769	0.588

Cap 3 chosen: best F1 and precision, and the product cost asymmetry (a false merge hides a story; a missed merge merely shows a duplicate) favors precision. Remaining errors at cap 3: 4 false merges (one is P32-type recurring roundup above the ceiling, which cosine alone cannot fix) and 4 missed merges (3 sit at cosine 0.8491–0.8499, just under the floor — the interleaving problem of §3.3 finding 1).

Evidence-boost below the threshold was tested and rejected: allowing merges in cosine [0.80, 0.85) with shared evidence would recover 2 true merges but introduce 4 false ones on the labeled set — the wrong trade under the cost asymmetry.

Production A/B (14-day window, 537 articles, `dedup_second_signal_20261012.txt`): Jaccard-only 491 clusters → +cosine 421 → +second signal **427**; only 4 gray-zone merges blocked, all plausibly-correct blocks (scope-mismatched weather/haze/road-project pairs). Effect on the general stream is small and protective because true same-story pairs usually share evidence anyway; the labeled hard stratum is where the precision gain materializes.

4. Reproduction

```
python scripts/eval_semantic_dedup.py # pair histograms + before/after clustering sim
```

5. Threats to validity / risks

1. **Heuristic labels** — resolved 2026-10-12: 40-pair human-labeled tuning set (§3.3). The labels confirmed 0.85 as the pragmatic operating point but exposed a systematic weak spot: same-location *recurring* events (daily haze readings, same-road crashes) false-merge at and above the threshold; a second gray-zone signal is the planned fix.
2. **False merges hide distinct stories** (see §3.2): ~23% at 0.85 by manual sample. Mitigations if this hurts in practice: raise the threshold (cross-lingual catches start dropping above ~0.89) or require an additional weak signal (shared distinctive token) alongside cosine.
3. **Fresh articles may lack embeddings** when Ollama was down at prefetch time — they silently get Jaccard-only treatment until embedded (observed: 7/1,746 unembedded right after a prefetch cycle).
4. **Batch composition matters**: the 312→249 figure is one 14-day global batch; per-user digest batches are smaller, so the delivered-item reduction will be smaller in absolute terms (though the same *rate* applies to every digest).
5. Embedding quality still bounds detection: same-story pairs with very different framings can sit under 0.85 (floor of `jaccard_caught` is 0.755) — those remain for the Jaccard/body layer or slip through.